Voorhees talk on exceptional cases. Nov 26, '56

Possible Indicators: I, ∈, Z?, DC, C?
Z? ↑ zero mantissa ?
C? ↑ cancellation ?

Exp. ovs & uns only when at least one of the operand mantissas is non-zero.

Notation: 0 indicates zero mantissa.
η " non-zero mantissa

(1) If I or ∈ exists in either operand, the primary opn. (+, −, ×, ÷) is not performed. (norm? maybe)

(2) If one or both of the operands is 0 the primary opn is not performed. (norm?)

Four types of nos.

1. η ordinary no.
2. 0 zero mantissa no, not the result of underflow, 0
3. $\eta_I$ = I ord. no. with I bit
4. $0_\epsilon$ zero mant. with underflow tag.

others: $\eta_\epsilon$ is set to $0_\epsilon$ in any operation
$0_I$ no opn. can produce this.
$\eta_{I\epsilon}$ do not exist
$0_{I\epsilon}$ " " "

assume division is normalized

Breakins: 4 triggers (either can be inhibited)
[Hi OV] 1 [Lo OV] 3 [DC] 5
[Hi UN] 2 [Lo UN] 4

1. Triggers are always turned on by exceptional events
2. 2 bits in opn. needed. (1 for breakin or no breakin) (end)
(1 for reset or no reset, triggers) (beginning)

3, (a) Turn on corresp. triggers when exceptional case occurs.
(b) Loc. of instr. is preserved,
(c) Test for BI or no BI
(1) If BI tr to fast mem. loc. 1, 2, 3, 4 or least no. in 2-trigger on (do not reset) & store loc. of BI instr.

tags may be on stores only?

(2) If no BT leave trigger on, leave loc. preserved

@Reset.

4. Reset, reset all 4 triggers & clear loc. preserve, done before performing primary instr.

+, −

for normalized

| | $n$ | | $n_I$ | | $\mathcal{O}$ | | $\mathcal{O}_e$ | |
|---|---|---|---|---|---|---|---|---|
| | BI | noBI | BI | noBI | BI | noBI | BI | noBI |
| $n$ | = $n$ = $\mathcal{O}$ unnorm. A+B ($\mathcal{O}_e$ cannot norm) Result exp. $n_I$ set | | cannot BI | ± $n_I$ | unnorm result | $n$ or $\mathcal{O}$ ($\mathcal{O}_e$ due to norm) | unnorm result | $n$ or ($\mathcal{O}_e$ due to norm) |
| $n_I$ | | | cannot BI | $n_I$ | cannot BI | ± $n_I$ | cannot BI | ± $n_I$ |
| $\mathcal{O}$ | | | | | Cannot BI | $\mathcal{O}$ | cannot BI | $\mathcal{O}_e$ (or $\mathcal{O}$?) |
| $\mathcal{O}_e$ | | | | | | | Cannot BI | $\mathcal{O}_e$ |

change due to unnorm, could use reserved $a \mid 0 = 0$ case

mpy

| | $\eta$ BI | $\eta$ noBI | $\eta_I$ | $\Theta$ | $\Theta_\epsilon$ |
|---|---|---|---|---|---|
| $\eta$ | trunc? $\eta$ / $\eta_I$ / $\Theta_\epsilon$ due to exp / ? / ermom | $\eta$ / $\Theta_\epsilon$ due to norm | C | C \| $\Theta$ | C \| $\Theta_\epsilon$ ($\alpha\Theta$?) |
| $\eta_I$ | | | C | C \| $\Theta$ ($\eta_I$?) | C \| $\Theta_\epsilon$ ($\eta_I$?) |
| $\Theta$ | | | | C \| $\Theta$ | C \| $\Theta_\epsilon$ ($\alpha\Theta$?) |
| $\Theta_\epsilon$ | | | | | C \| $\Theta_\epsilon$ ($\alpha\Theta$?) |

divide

| | $\eta$ | | $\eta_I$ | | $\Theta$ (numerator) | | $\Theta_\epsilon$ | |
|---|---|---|---|---|---|---|---|---|
| $\eta$ | — / ? / ? / $\Theta_\epsilon$ exp / $\Theta_\epsilon$ norm | $\eta$ / $\eta_I$ / $\Theta_\epsilon$ exp / $\Theta_\epsilon$ norm | C | $\eta_I$ | C \| $\Theta$ | | C | $\Theta$ ($\alpha\Theta_\epsilon$) |
| $\eta_I$ | C | $\Theta_\epsilon$ ($\Theta$) | C | $\eta_I$ | C \| $\Theta$ | | C | $\Theta$ ($\Theta_\epsilon$) |
| | DC | noDC | DC | noDC | DC | noDC | DC | noDC |
| $\Theta$ | A', B' | $\eta_I$ | A' B' | $\eta_I$ | A' B' | $\Theta$? | A' B' | $\eta_I$? $\Theta_\epsilon$? |
| $\Theta_\epsilon$ | A', B' | $\eta_I$ | | $\eta_I$ | | $\Theta$? $\eta_I$ | | |
| $\eta$ | ~~0 1~~ | $\eta_I$? | | | | | | |

(denom)

Signed Exp:

1) $\Theta$: $[-(2^{12}-1), \pm\Theta]$ , $\Theta_\epsilon$: $\Theta$ with $\epsilon$, $\eta_I$: $I$ or $\pm$?

2) $\Theta$: [any exp, $\pm\Theta$] , $\Theta_\epsilon$: $\Theta$ with $\epsilon$ ,

mod exp.

(1) $\Theta$: $[\pm\Theta, \pm\Theta]$

or [any, $\pm\Theta$]